

The WHY and HOW of visualization

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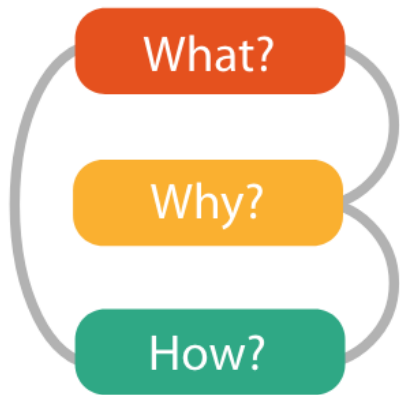
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- 2 How to **visually encode** the data?
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The "what - why - how" approach (Munzner, 2014)



- **What** : What data is shown to the user?
- **Why** : Why do the users use the visualization technique?
- **How** : How is the data encoded and what interactions are supported?

Why to **use** a visualization?

Action: to **analyze** the data

➔ Analyze

➔ Consume

➔ Discover



➔ Present



➔ Enjoy



➔ Produce

➔ Annotate



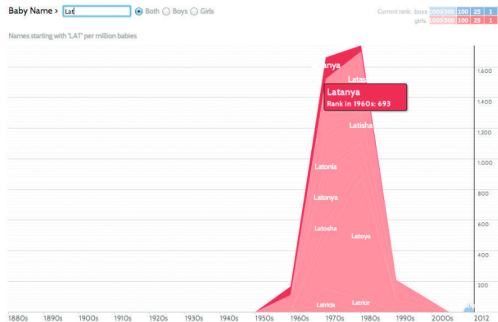
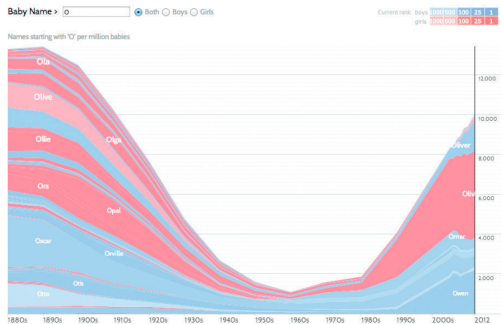
➔ Record



➔ Derive



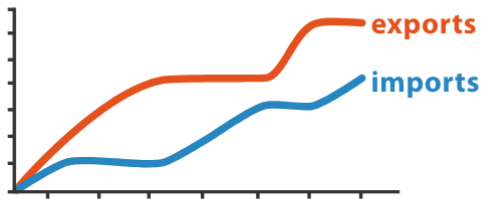
Example: Consuming data for entertainment



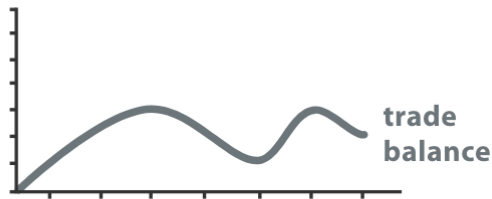
Name Voyager : **designed for parents** searching for a baby name → **used by non-parents** to analyze the evolution of baby names

Source : www.bewitched.com/namevoyager.html

Example: Producing new information



Original Data



$$\text{trade balance} = \text{exports} - \text{imports}$$

Derived Data

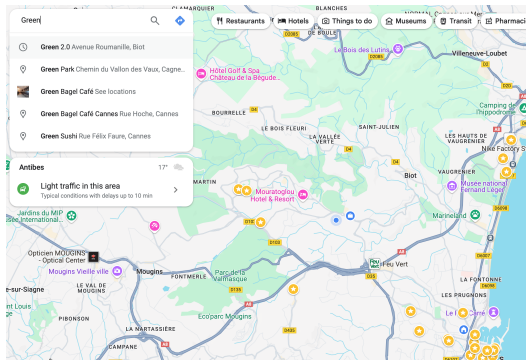
Action: to **search** in the data

➔ Search

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

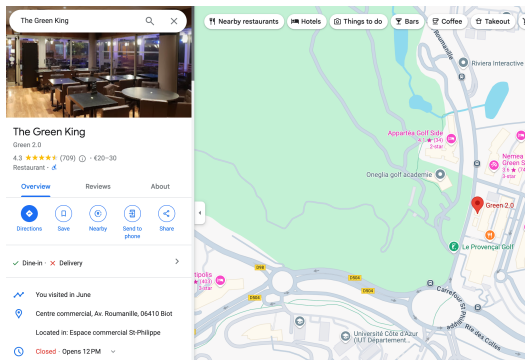
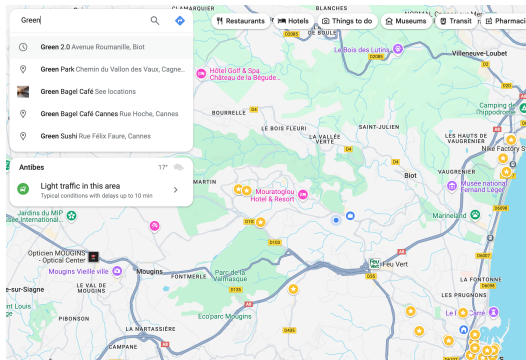
Lookup: example on Google Maps

Task: Search for informations about the restaurant **Green 2.0** located in **Saint-Philippe**.



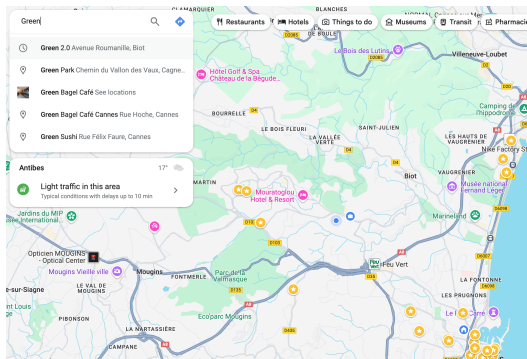
Lookup: example on Google Maps

Task: Search for informations about the restaurant **Green 2.0** located in **Saint-Philippe**.



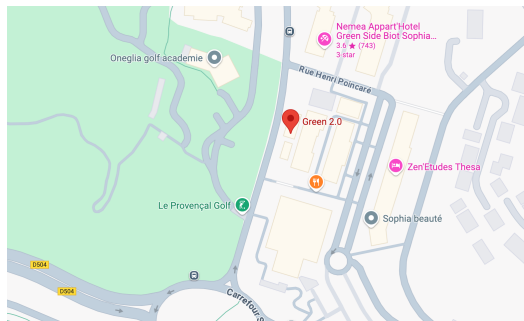
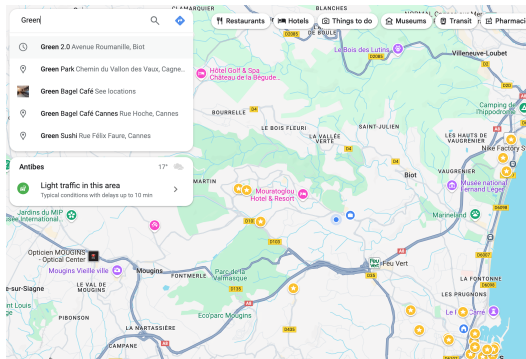
Locate: example on Google Maps

Task: Where is the **Green 2.0** restaurant located?



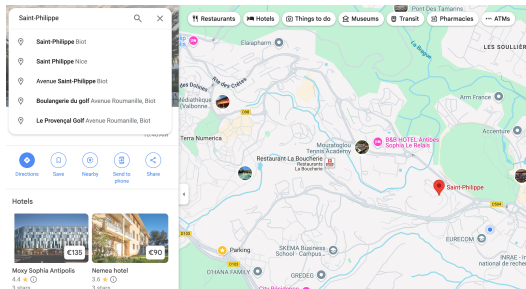
Locate: example on Google Maps

Task: Where is the **Green 2.0** restaurant located?



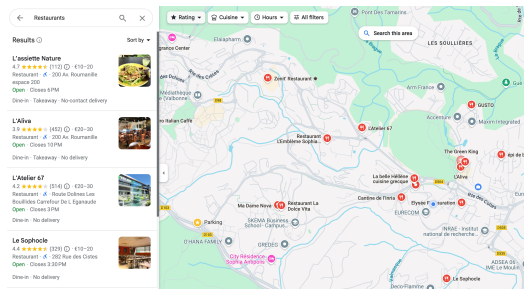
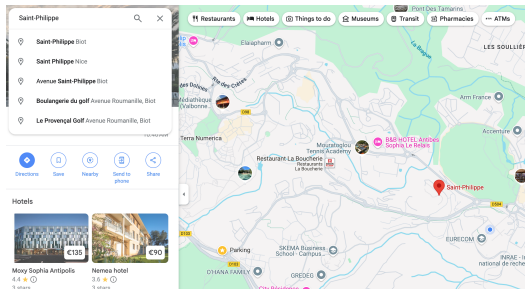
Browse: example on Google Maps

Task: What restaurants can be found in **Saint-Philippe**?



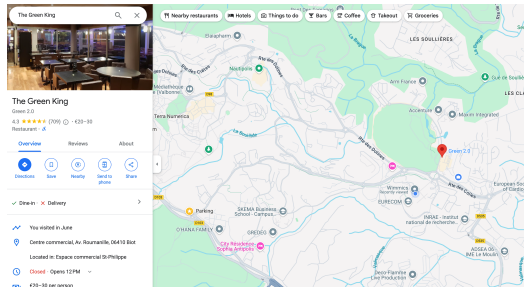
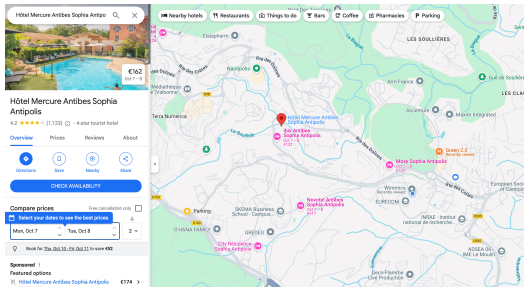
Browse: example on Google Maps

Task: What restaurants can be found in **Saint-Philippe**?



Explore: example on Google Maps

Task: Discover whatever is available on the map.



Action : to **query** the data

After finding the targets, the user can **query** them

➔ Query

➔ Identify



➔ Compare



➔ Summarize



Targets

➔ All Data

➔ Trends



➔ Outliers



➔ Features



➔ Attributes

➔ One

➔ Distribution



➔ Extremes



➔ Many

➔ Dependency



➔ Correlation

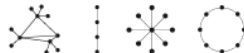


➔ Similarity



➔ Network Data

➔ Topology



➔ Paths



➔ Spatial Data

➔ Shape



How to **visually encode** the data?

Graphical Elements: spatial dimensions

Points (0D) | Lines (1D) | Area (2D) | Volume (3D)

➔ Points



➔ Lines



➔ Areas



Visual channels: change the appearance of marks

➞ Position

➞ Horizontal



➞ Vertical



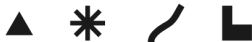
➞ Both



➞ Color



➞ Shape



➞ Tilt

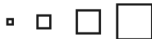


➞ Size

➞ Length



➞ Area



➞ Volume



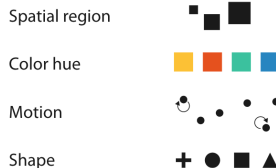
The effectiveness of visual channels

Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude Channels: Ordered Attributes**



➔ **Identity Channels: Categorical Attributes**



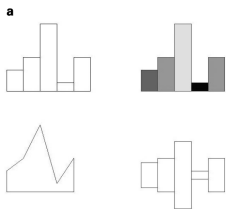
Glyphs: a special kind of mark

Glyph: a compact visual symbol that encodes multiple data attributes into a single graphical mark

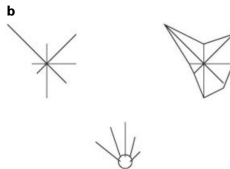
Key characteristics:

- **Multivariate representation:** Each part of the glyph (shape, color, size, orientation, texture, etc.) maps to a different data dimension.
- **Local visualization:** Glyphs represent data at the item level, making them suitable for comparing individual records.
- **High information density:** They allow many attributes to be shown simultaneously within limited space.
- **Flexible design:** Glyphs can be tailored to the domain (e.g., medical, scientific, geographical).

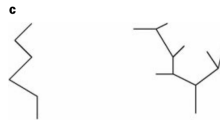
Examples of glyphs



Variations on Profile glyphs



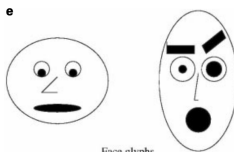
Stars and Anderson/metroglyphs



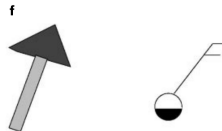
Sticks and Trees



Autoglyph and box glyph



Face glyphs



Arrows and Weathervanes

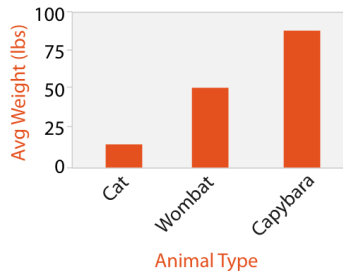
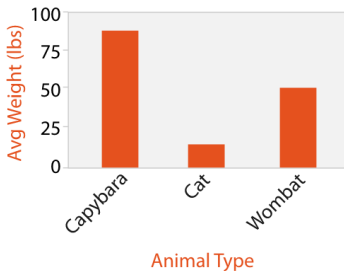
Source: [Ward \(2002\)](#)

How to choose the **right** visualization technique?

Functions of Visualization (Ribecca, 2014)

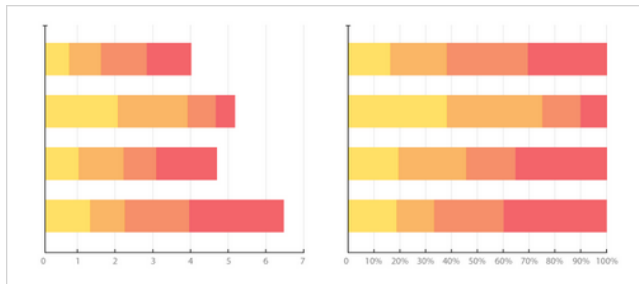
- **Comparison**: highlight differences or similarities
- **Proportion**: show part-to-whole relationships
- **Relationships**: reveal connections or correlations
- **Hierarchy**: present ordered or ranked structures
- **Location**: map data to geographic regions
- **Distribution**: show frequency or spread
- **Movement/Flow**: depict movement or transfer
- **Patterns**: uncover recurring forms or structures
- **Time**: show changes and trends over time

Bar Chart: used for **Comparisons**



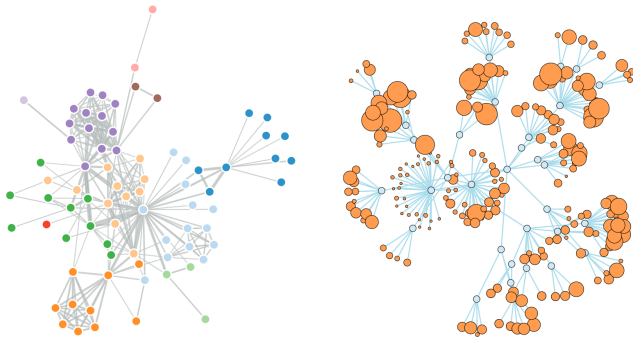
- **Data:** 1 quantitative and 1 categorical attribute
- **Marks:** lines
- **Visual channels:** horizontal position (categorical attribute), vertical position (quantitative attribute), and size (line height)
- **Other functions:** patterns
- **Scalability:** up to a hundred levels for the categorical attribute

Stacked Bar Chart: used for **Proportions**



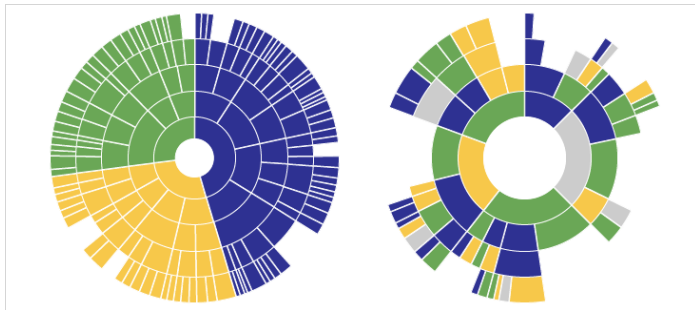
- **Data:** 1 quantitative and 2 categorical attributes
- **Marks:** lines
- **Visual channels:** horizontal position, vertical position, size (line height), and color (2nd categorical attribute)
- **Other functions:** comparisons, part-to-a-whole (100%)
- **Limitations:** harder to read with many colored segments; side-by-side comparison suffers from the lack of a common baseline.

Node-link diagram: used for **Relationships**



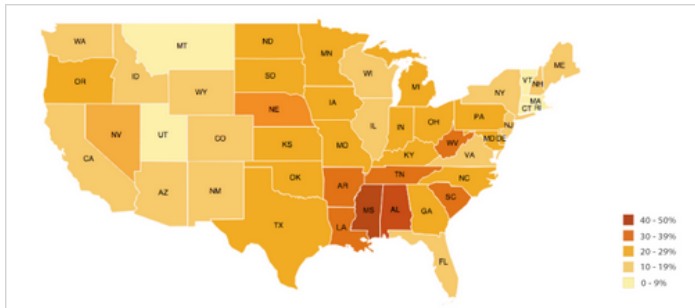
- **Data:** network (nodes and links), one quantitative and one categorical attribute
- **Marks:** dots and lines
- **Visual channels:** horizontal and vertical position, color, size (area and length)
- **Task:** explore relationships and paths between items
- **Scalability:** hundreds of nodes and links (density: $L < 4N$)

Sunburst: used for **Hierarchy**



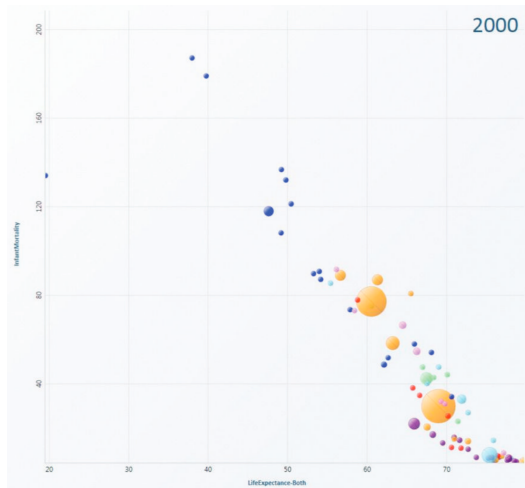
- **Data:** hierarchical data (1 quantitative and multiple categorical attributes – 1 per level)
- **Marks:** area
- **Visual channels:** tilt/angle (hierarchy), surface (quantitative), and color (categorical)
- **Other functions:** part-to-a-whole
- **Scalability:** moderate hierarchical depth (< 6 levels) and a few dozen nodes

Choropleth map: used for **Location**



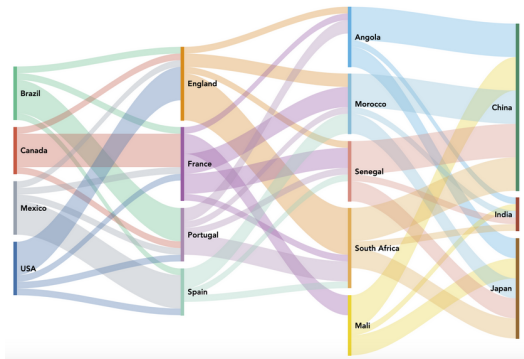
- **Data:** geospatial data, with a quantitative or categorical attribute
- **Marks:** area
- **Visual channels:** horizontal and vertical positions, color
- **Other functions:** comparisons, patterns
- **Limitations:** color is hard to judge; large regions dominate visually

Bubble Chart: used for **Distribution**



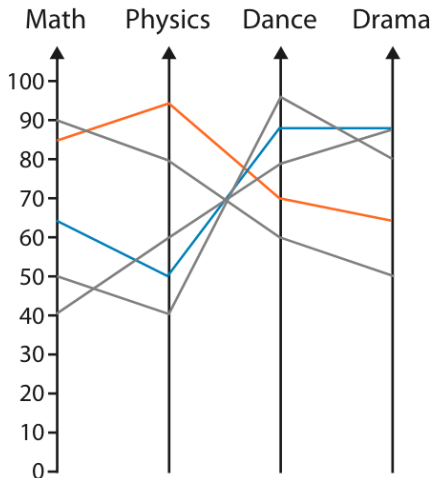
- **Data:** up to 3 quantitative and 1 categorical attributes
- **Marks:** points
- **Visual channels:** color (categorical), size (quantitative), horizontal and vertical position (quantitative)
- **Other functions:** comparisons, data over time, patterns, proportions, relationships
- **Scalability:** a few hundred items

Sankey Diagram: used for **Movement or Flow**



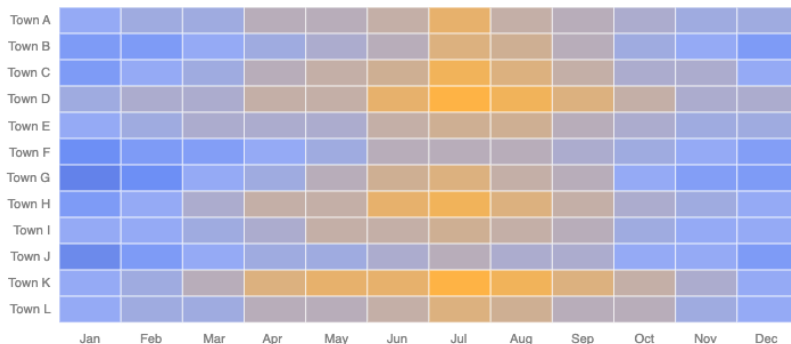
- **Data:** flows between categories; 1 quantitative (value) + a few categorical attributes
- **Marks:** nodes and links
- **Visual channels:** color (category), width (flow size), position (stage/group)
- **Other functions:** relationships, proportions, patterns, part-to-whole
- **Scalability:** tens of nodes, up to a few hundred links

Parallel Coordinates Plot: used for **Patterns**



- **Data:** multiple attributes (quantitative, categorical, temporal, etc)
- **Marks:** lines
- **Visual Channels:** horizontal and vertical positions, and color
- **Other functions:** comparisons, relationships
- **Scalability:** about a dozen attributes (axes) and hundreds of items

Heatmap: used for **Time**



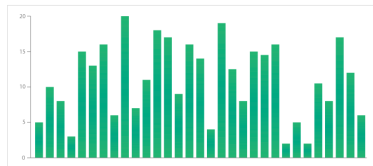
- **Data:** 2 categorical and one quantitative attribute
- **Marks:** area.
- **Visual channels:** color, horizontal and vertical position.
- **Other functions:** comparisons, patterns, relationships
- **Scalability:** millions of items with hundreds of levels for the categorical attribute.

Which technique should I use?

It depends on multiple factors:

- the **type and volume of data**
- the **number and type of attributes**
- the **function** you want to support:
comparison, patterns, relationships,
proportions, hierarchy, location,
distribution, movement or flow, data over
time, etc.

Bar Chart



Description

Also known as Bar Graph or Column Graph.

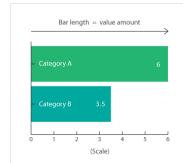
A Bar Chart uses either horizontal or vertical bars (column chart) to show discrete, numerical comparisons across categories. One axis of the chart shows the specific categories being compared and the other axis represents a discrete value scale.

Bar Charts are distinguished from Histograms, as they do not display continuous developments over an interval. Instead, Bar Chart's discrete data is categorical and therefore answers the question of "how many?" in each category.

Functions

Comparisons Patterns

Anatomy



The Visualization Catalogue : <https://datavizcatalogue.com/>

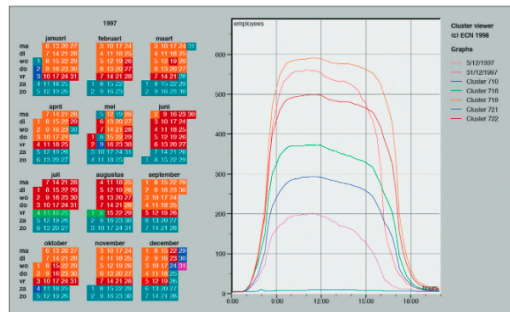
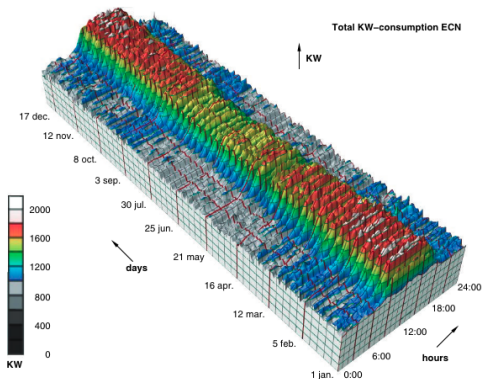
About the 3D

Use 3D only when it is **essential** for understanding the geometric structure of objects or scenes.

No unjustified 3D

- Positions are restricted to two dimensions (x-y plane), ignoring full 3D space.
- Depth (3D) positions are judged less accurately than planar (2D) ones.
- Occlusion causes information loss.
- Perspective distortion makes distant objects appear smaller and misplaces them visually.
- Inclined text is hard to read.

Exemple : Cluster-Calendar Time-Series Vis



Source: Van Wijk and Van Selow (1999)

How to **combine** multiple visualizations?

Multiple Views Systems or Dashboards

Using two or more distinct views to explore different aspects of a single conceptual entity.

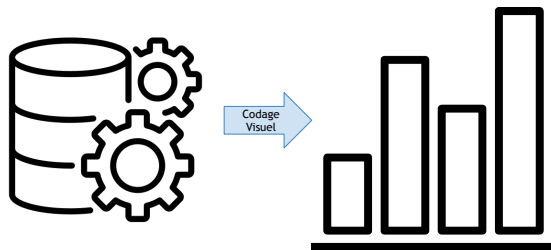


Source: [Wu et al. \(2015\)](#)

Advantages ([North and Shneiderman, 1997](#)) :

- Improve user performance
- Discover unexpected relationships
- Unify the desktop

Distinct views



Two **distinct** views → each view communicates a different aspect of the same conceptual entity:

- By presenting additional information
- By highlighting the same information

Examples of Different Datasets

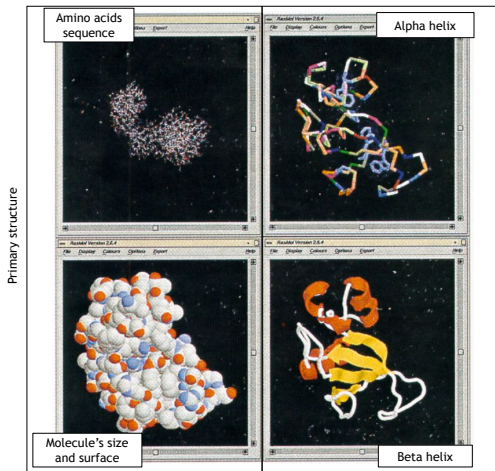
- One dataset may be a **subset** of another
 - **Example:** One view shows total revenue over time; another shows revenue for a specific company.
- One dataset may be **aggregated** from another
 - **Example:** One view shows average cost per restaurant type; another shows individual restaurant locations.
- Two datasets may be **completely different**
 - **Example:** One view shows people's movement in a city; another shows the function of visited places.

When to Use Multiple Views (Baldonado et al., 2000)

- **Diversity**: There is a variety of attributes, models, user profiles, abstraction levels, and genres.
- **Complementarity**: Distinct views highlight correlations or differences.
- **Decomposition**: Split complex data into multiple views to ease exploration.
- **Parsimony**: Use multiple views efficiently and minimally.

- **Attributes**: histogram showing the distribution of images by creation date or colors used
- **Models**: conceptual representation or architecture of a complex system
- **User Profiles**: different preferences, expertise levels, roles, etc.
- **Levels of Abstraction**: detailed road map versus an overview of the metropolitan region
- **Genres**: block diagram or pseudo-code representation within a software module

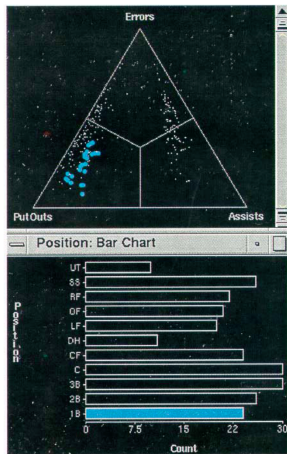
Complementarity



- Highlight correlations and differences
- Simplify comparisons
- Clarify complex data

Chemistry of the barnase molecule. Source: [Baldonado et al. \(2000\)](#).

Decomposition



Source : [Baldonado et al. \(2000\)](#)

- Highlight relationships between different data dimensions
- "Divide and conquer" approach: reduces the amount of data analyzed at once

Parsimony

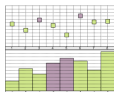
Despite the benefits of using multiple views:

- They increase the user's cognitive load
- They take up screen space
- They may require learning new concepts
- Introduce additional views only for compelling reasons
- Merge similar views whenever possible

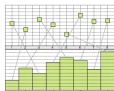
How to Use Multiple Views

- **Resource Optimization:** Balance spatial and temporal costs
- **Self-evidence:** Use perceptual cues to convey relationships between views
- **Consistency:** Ensure coherence and uniformity across visual elements and interactions
- **Attention Management:** Guide the user's focus to the relevant view using perceptual cues

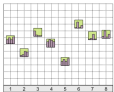
Dashboard Layout (Javed and Elmqvist, 2012)



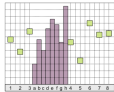
Juxtaposition: place views side-by-side



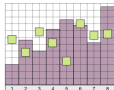
Integration: side-by-side placement with visual links



Superimposition: superimpose two visualizations

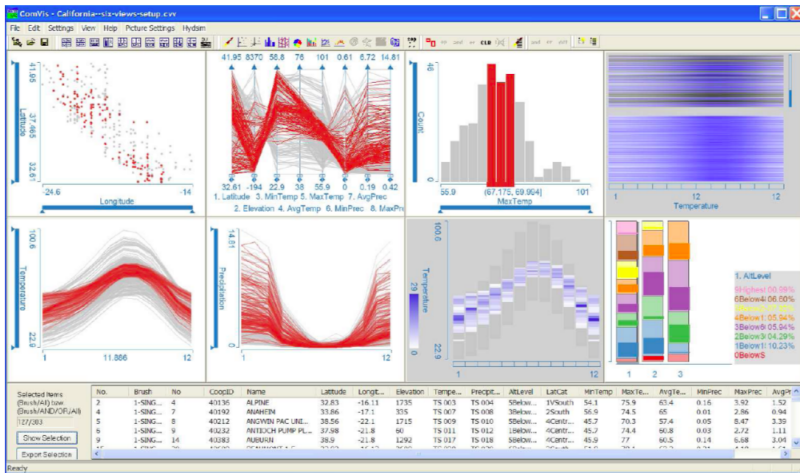


Overload: use the space of one visualization for another



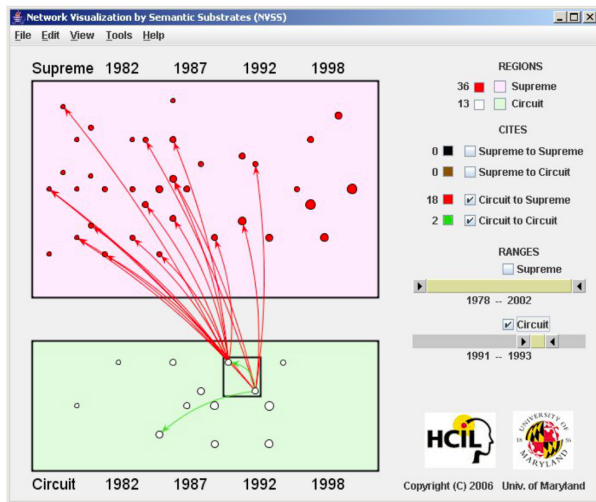
Nesting: place the content of one view inside another

Example of Juxtaposed Views



Meteorological dashboard. Source : Javed and Elmqvist (2012)

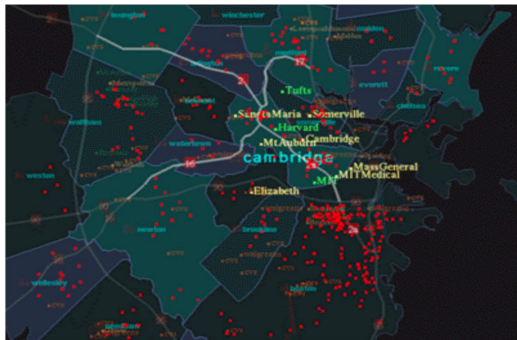
Examples of Integrated Views



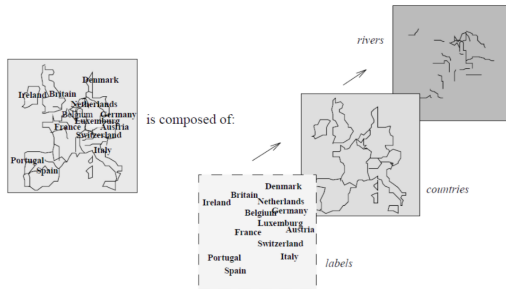
- Better relational cognition
- More visual disorder

Judicial System Visualization. Source : (Javed and Elmqvist, 2012).

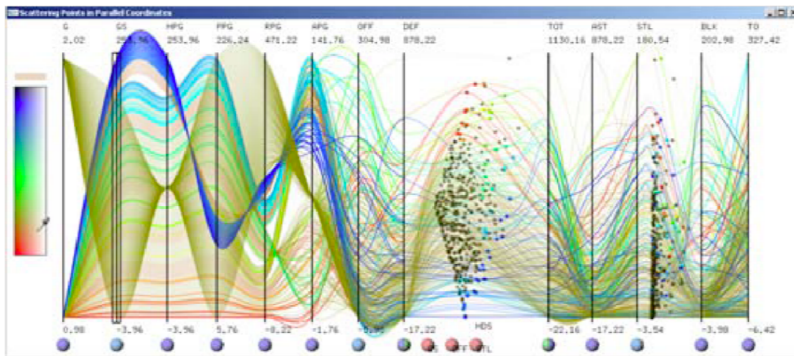
Examples of Superimposed Views



Geospatial visualization of crime data. Source : [Javed and Elmqvist \(2012\)](#)



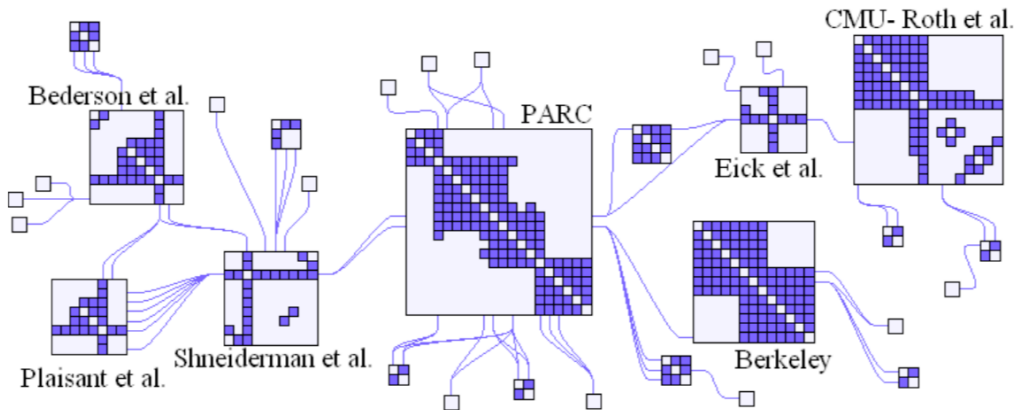
Example of Overloaded Views



Overlapping points between two axes in a parallel coordinates plot. Source : [Javed and Elmquist \(2012\)](#)

- Similar to superimposed views
- **Difference:** No spatial link between the visualizations

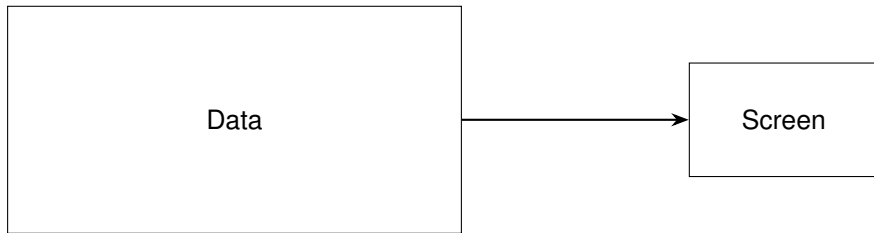
Example of nested views



Co-authorship network exploration. Source : [Javed and Elmqvist \(2012\)](#)

Why and how to **interact** with the visualization?

Why interact?



- **Support users** in analytical tasks, especially query-related actions.
- Enable exploration of increasingly **large and high-dimensional data**.
- Allow **filtering, focusing, and refinement** of visible information.
- Provide **details on demand** without cluttering the display.
- Let users **change perspectives** (zoom, pan, reorder) to reveal patterns.

Data Selection

Click-based selection

The image shows a mobile application interface for hotel booking. On the left, a vertical panel displays details for the 'Nemea hotel'. At the top, there's a photo of the hotel building. Below it, the name 'Nemea hotel' is followed by 'Nemea Appart'Hotel Green Side Biot Sophia Antipolis', a 3.6-star rating, and '(743)' reviews. It's labeled as a '3-star tourist residence'. There are tabs for 'Overview', 'Prices', 'Reviews', and 'About'. Below these are icons for 'Directions', 'Save', 'Nearby', and 'Share'. A blue button says 'CHECK AVAILABILITY'. Further down, a 'Compare prices' section shows check-in/out dates (Mon, Oct 7 to Tue, Oct 8) and a price of €20. Below that, 'Sponsored' and 'Featured options' list the 'Nemea hotel' for €74. 'All options' also lists the 'Nemea hotel' for €74 and 'Weekendesk.fr' for €90. On the right, a map shows the location of the hotel with a red pin. A box labeled 'Elément sélectionné' points to this pin. Other nearby locations like 'Moxy Sophia Antipolis', 'Le Provençal Golf', and 'Basic-Fit Biot' are also visible on the map.

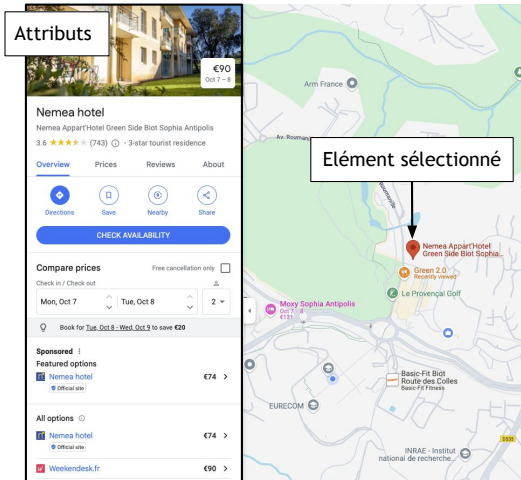
Attributs

Elément sélectionné

Data Selection

Click-based selection

Attributs



Nemea hotel

Nemea Appart'Hotel Green Side Biot Sophia Antipolis

3.6 ★★★★★ (743) · 3-star tourist residence

Overview Prices Reviews About

Directions Save Nearby Share

CHECK AVAILABILITY

Compare prices

Free cancellation only ☐

Check in / Check out

Mon, Oct 7 Tue, Oct 8 2

Book for Tue, Oct 8 - Wed, Oct 9 to save €20

Sponsored :

Featured options

Nemea hotel €74 >

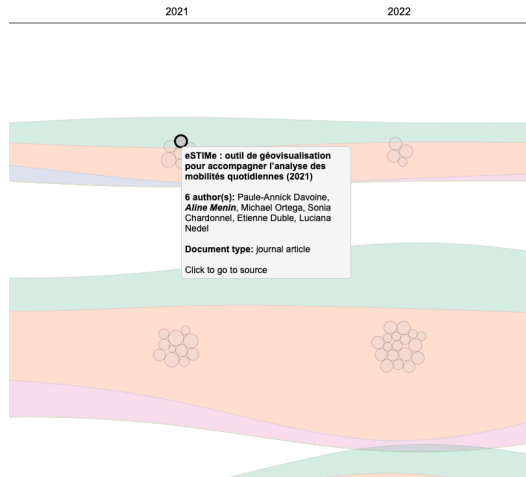
All options

Nemea hotel €74 >

Weekendesk.fr €90 >

Elément sélectionné

Hover-based selection (tooltip)



Item reordering

Ascending order of income

Product Category	Revenue Won	Qualified P...	Forecast %
Warranties	\$3,271.66K	\$6,259.22K	136%
1 Year Warran...	\$3,271.66K	\$6,259.22K	136%
Devices	\$3,672.98K	\$6,605.71K	128%
E-reader Stan...	\$137.48K	\$295.13K	144%
E-reader Dia...	\$156.58K	\$361.14K	129%
E-reader Plati...	\$554.24K	\$931.40K	149%
E-reader Dia...	\$971.02K	\$1,562.41K	127%
E-reader Plati...	\$1,853.66K	\$3,455.63K	133%
Accessories	\$4,485.19K	\$7,035.43K	128%
Black cover 6"	\$1,463.81K	\$2,058.70K	117%
E-reader pen	\$1,509.29K	\$2,717.23K	141%
Black cover 7"	\$1,512.09K	\$2,259.50K	126%
Total	\$11,429.83K	\$19,900.36K	136%

Item reordering

Ascending order of income

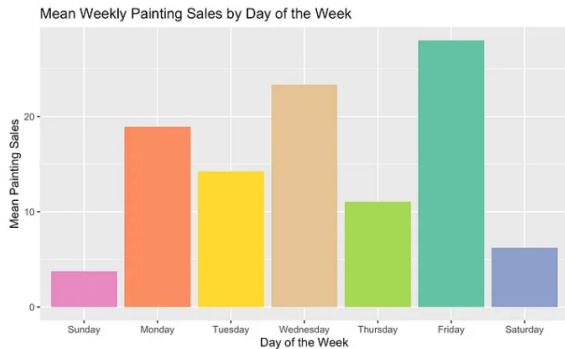
Product Category	Revenue Won	Qualified P...	Forecast %
Warranties	\$3,271.66K	\$6,259.22K	136%
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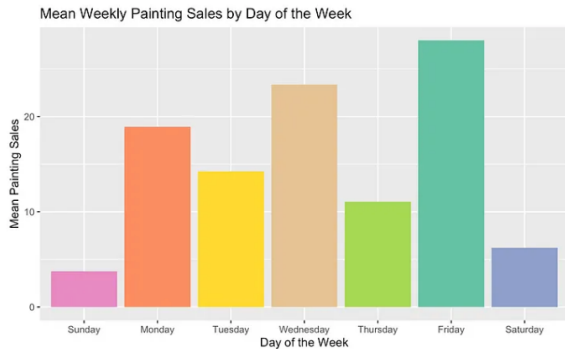
Reordering items

Ordered by day of the week (natural order)

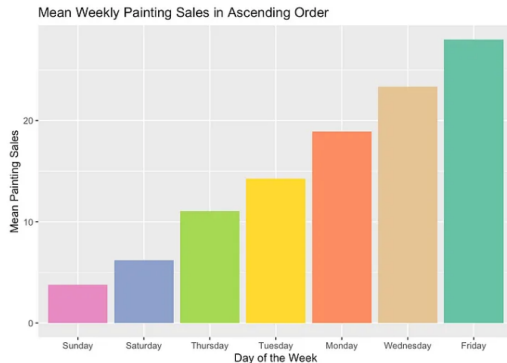


Reordering items

Ordered by day of the week (natural order)



Ascending order of sales (comparison)



Scalability: interaction strategies

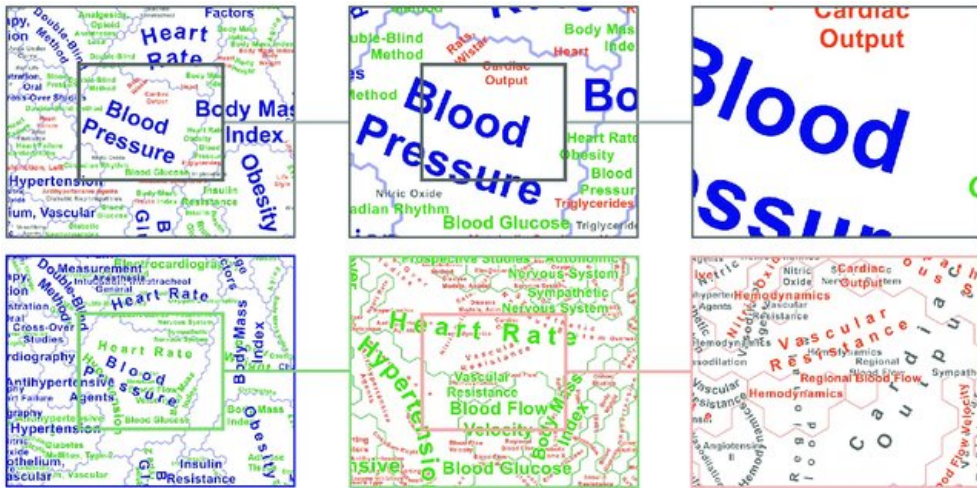
Information compression:

- **Reducing the size of graphical elements (geometric zoom):** Adjust visual element sizes to match the zoom level while preserving proportions.

Reducing the amount of information:

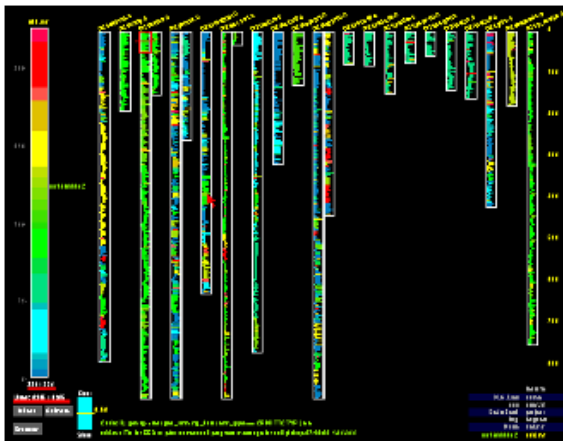
- **Lossless compression (semantic zoom):** The representation adapts to the zoom level without losing content.
- **Increasing density:** Maximize the amount of information displayed within the available space.

Geometric Zoom vs. Semantic Zoom



Source: Skupin et al. (2013)

Example of increasing density



Filtering system: reducing the number of displayed elements

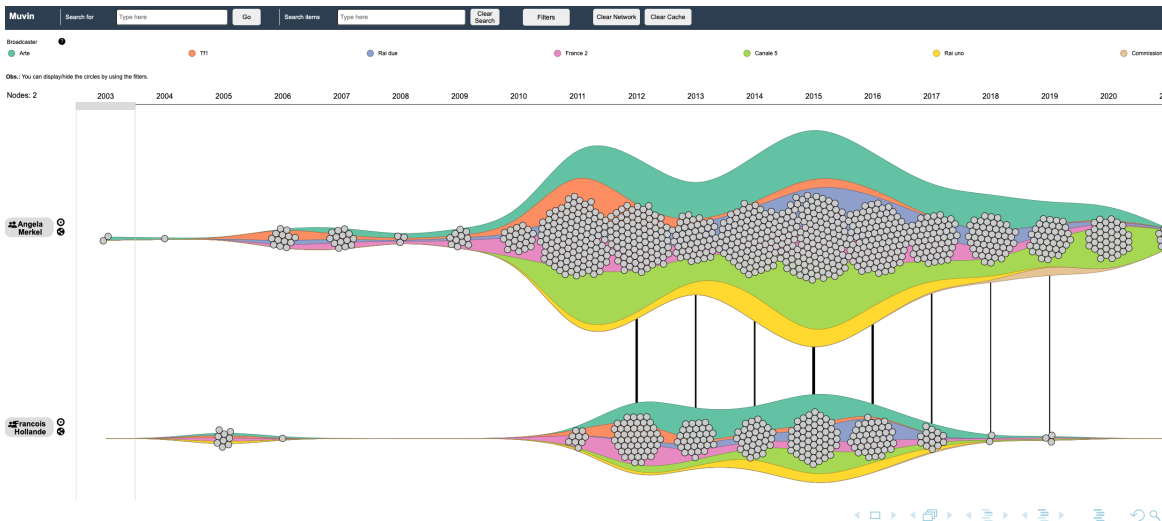
- **Item filtering:** decreases the number of displayed items while keeping the same set of attributes.
- **Attribute filtering:** keeps the same number of visible items but reduces the number of attributes shown for each one.

Advantages:

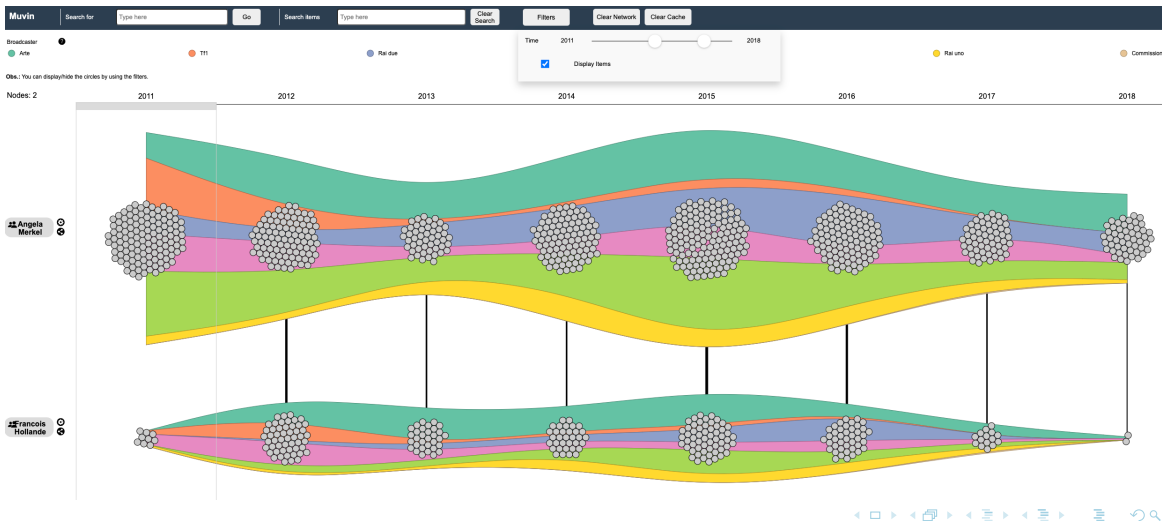
- Focus on specific, relevant information.
- Simplifies the data, making it more accessible and meaningful to the user.
- Enables a personalized experience tailored to analytical needs.

Drawback: Loss of information

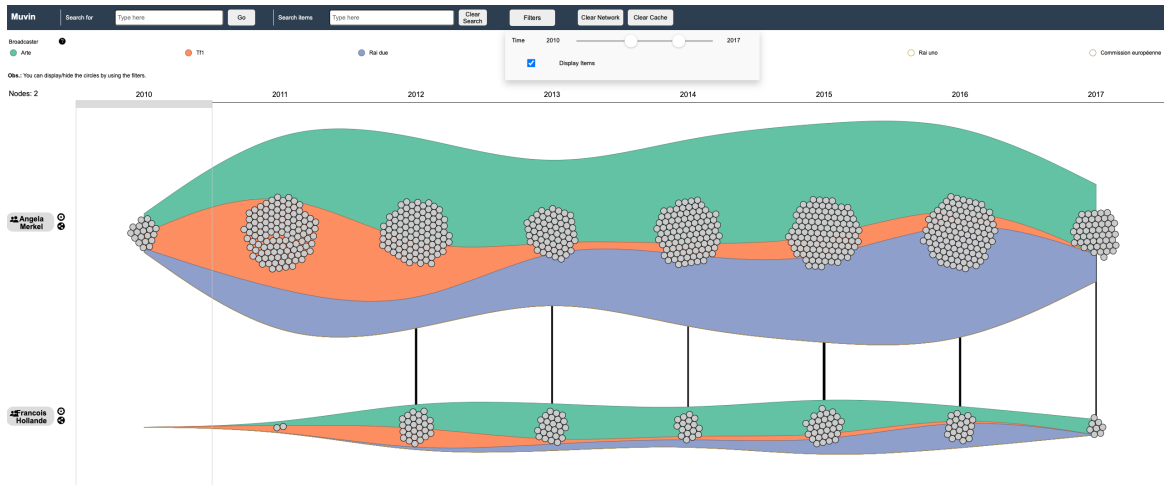
Item filtering based on **categorical** attribute



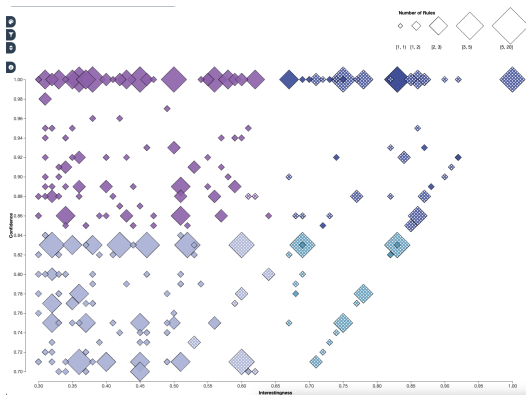
Item filtering based on **temporal** attribute



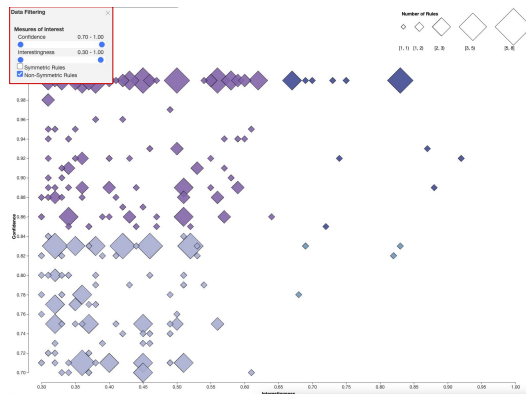
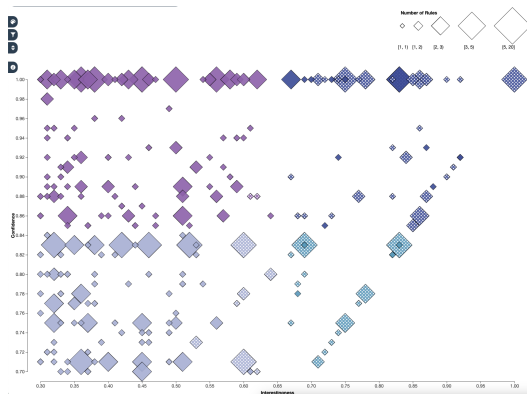
Item filtering through **multiple** filters



Item filtering based on **quantitative** attribute

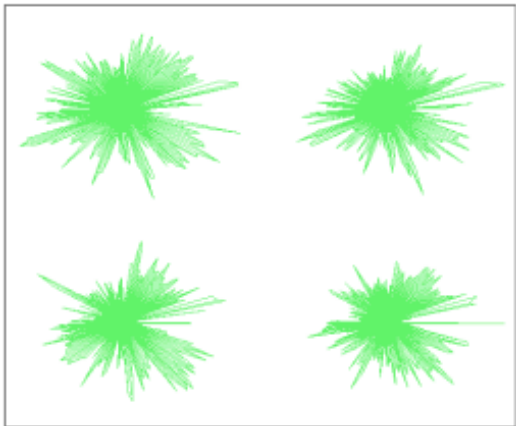


Item filtering based on **quantitative** attribute



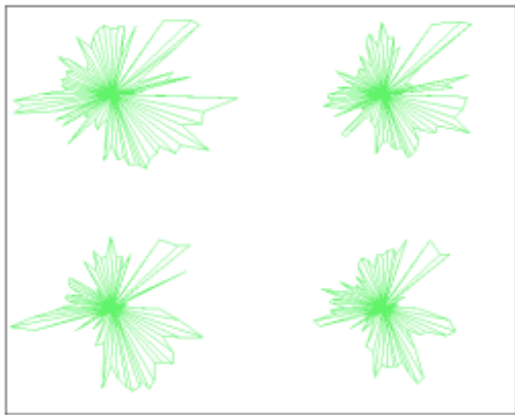
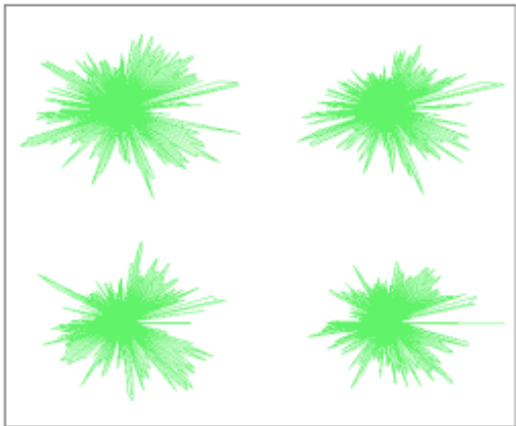
Attribute filtering: example on DOSFA (Yang et al., 2003)

Glyphs representing medical data across 215 dimensions



Attribute filtering: example on DOSFA (Yang et al., 2003)

Glyphs representing medical data across 215 dimensions



Data Aggregation

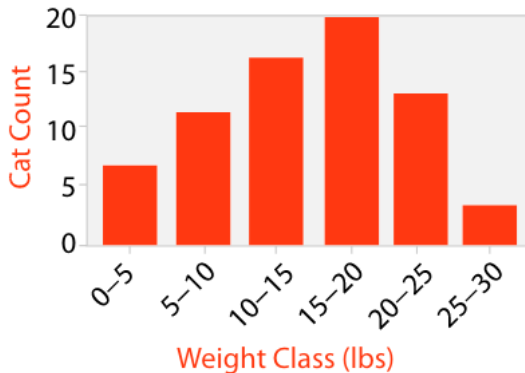
Derive an item or a group of items that represents a larger set of items

- What criteria can be used for aggregation?
 - Group by category
 - Spatial partitioning
 - Statistically (clustering)
 - Arbitrarily defined by the user
- What values are associated with an aggregation?
 - Mathematical functions: count, average, min, max
 - Semantic abstraction
- Multi-level grouping → hierarchies / trees

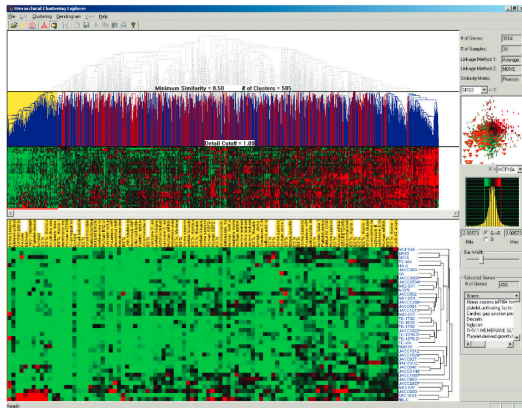
Example of data aggregation

Distribution of cat weights in a neighborhood, grouped in 5-pound bins:

- 1 Divide the range of the original attribute into classes (bins).
- 2 Count and record the number of items in each class as a derived, ordered attribute.



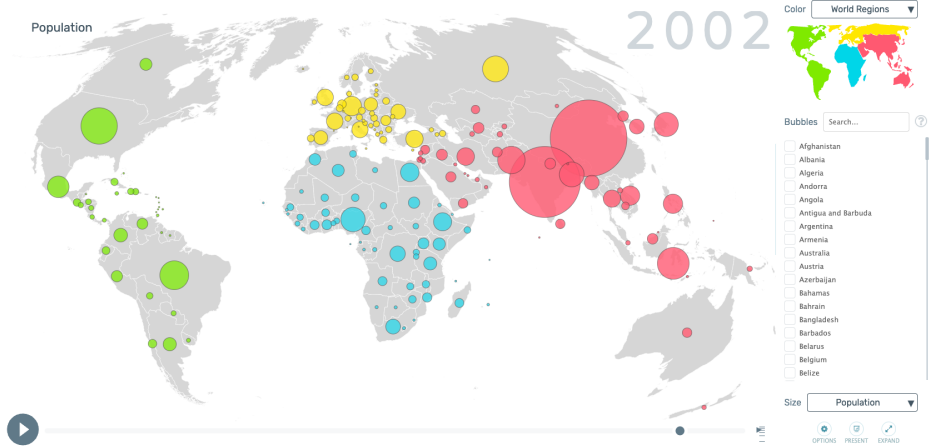
Overview + Details



Show the full dataset while allowing details on demand.

- No distortion
- Overview and details can be seen simultaneously
- **Drawback:** users must consciously shift their focus

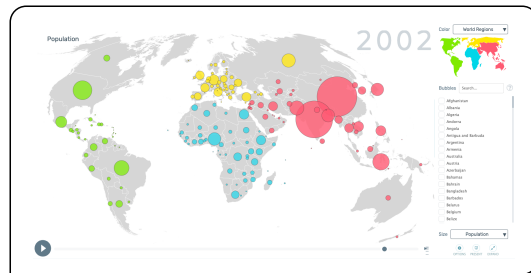
Vue de l'ensemble



Gapminder : <https://www.gapminder.org/tools/>

Overview

- **Graph elements:**
 - Spatial orientation through the geographic map
 - Informative context
 - Presence or absence of data
- **Tasks:** identifying trends
- Direct access to information
- Rapid search and exploration



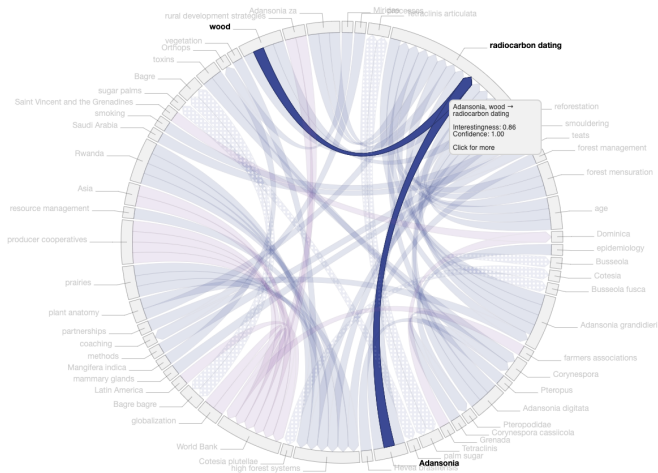
Details and focus



- The result of a data processing workflow
- Details on a subset of the data
- Geometric zoom



Details and focus on a chord diagram

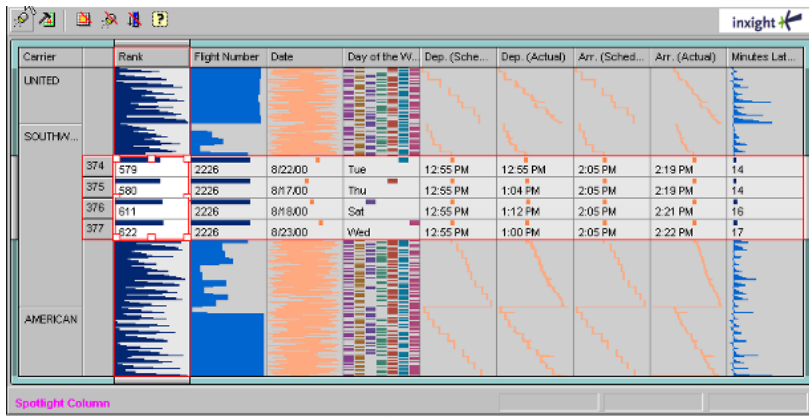


Try it here : <http://dataviz.i3s.unice.fr/arviz/issa>

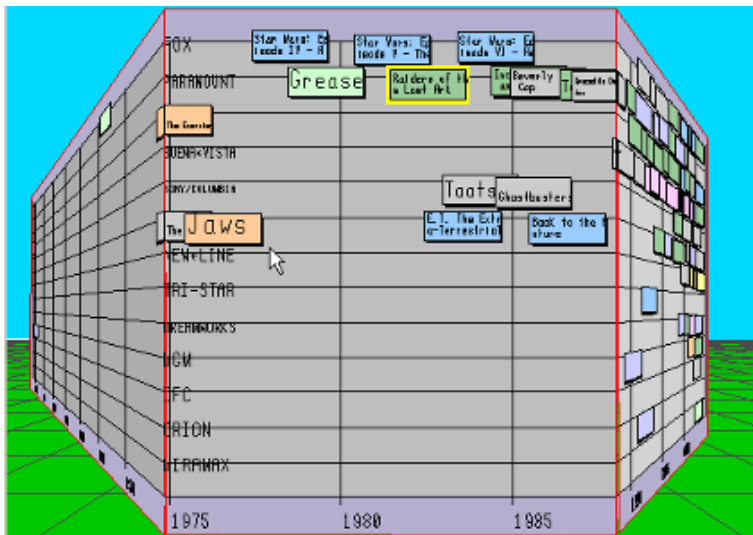
Examine a specific **region of interest (focus)** in detail while keeping the **surrounding data (context)** visible.

- A single view presents information in context.
- Contextual information is displayed near the point of interest.
- Detailed exploration without losing awareness of the overall dataset.
- **Drawbacks of distortion:**
 - Some areas may become difficult to interpret.
 - The data structure can be obscured.

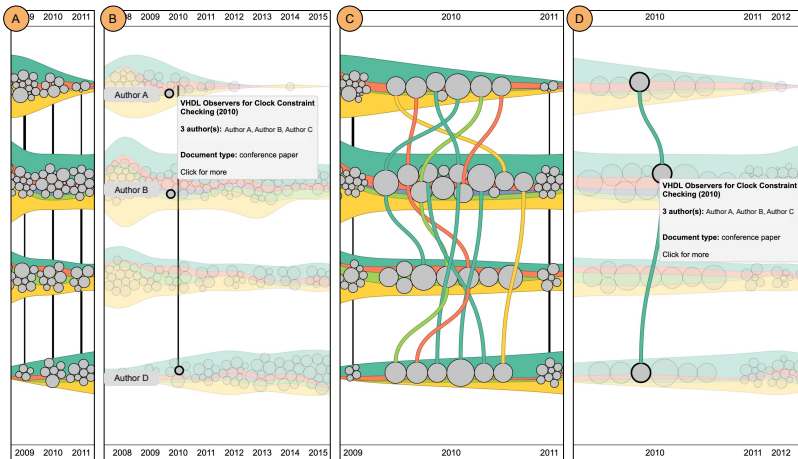
Focus + Context: example **without** distortion



Focus + Context: example **with** distortion

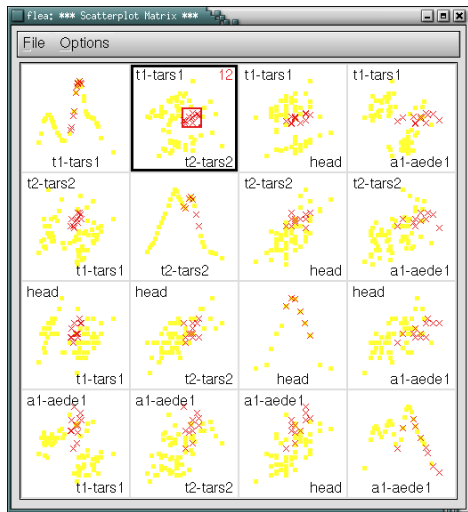


Focus + context + details



Try it here: <https://dataviz.i3s.unice.fr/muvin/hal>

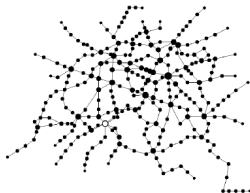
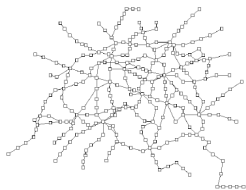
Brushing & Linking



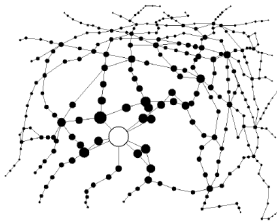
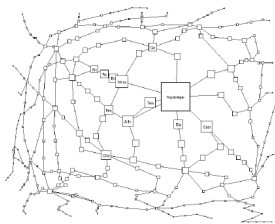
Selecting data in one view and **highlighting** it across coordinated views to explore relationships.

- Combine complementary visualization techniques
- Interactive changes propagate across all views
- Show the behavior of points or groups in each chart
- Highlight selected points

Fisheye lenses



A distortion technique that **enlarges** a region of interest **while compressing** surrounding areas, **preserving context**.

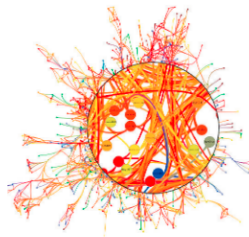


- Show local details and global context simultaneously
- Highlight relationships between a focused item and its neighbors

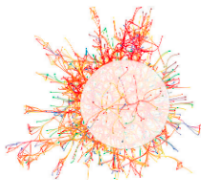
Fisheye + details



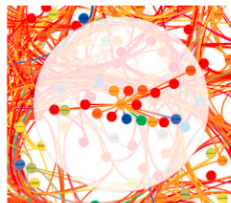
(a)



(b)



(c)



(d)

Rules of Thumb

- Avoid unjustified 3D
- Avoid unnecessary 2D → if a simple query answers the question, visualization may not be needed
- Rely on visual perception rather than users' memory
- Resolution before immersion: pixels are precious
- Responsiveness is essential
- Ensure it works in black and white
- Function first, form second

Shneidermann's mantra

"Overview first, zoom and filter, then details-on-demand"

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